

Floating Point Representation

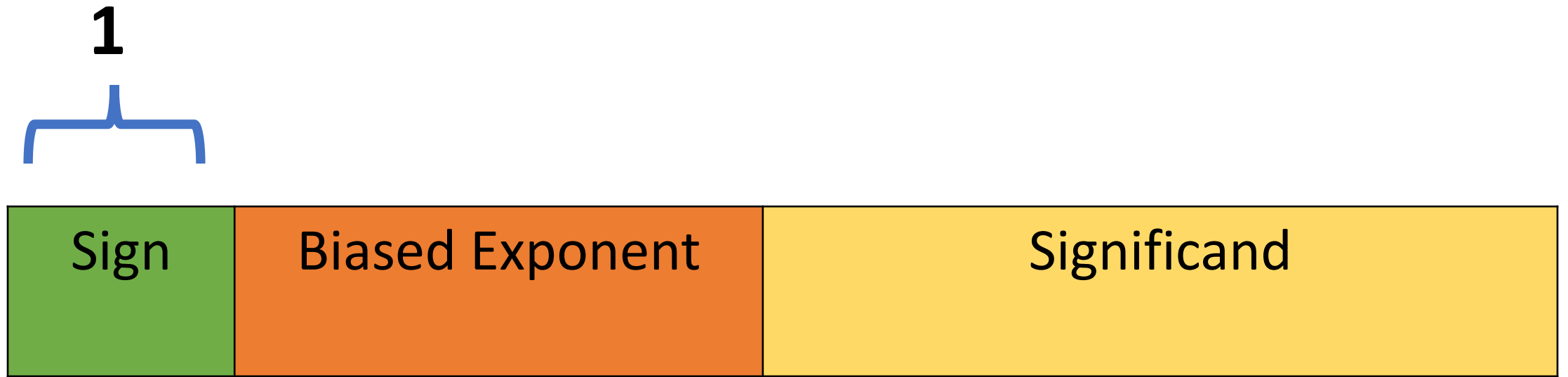
Context

- General floating point representation
- IEEE - 754 single precision and double precision floating point representation
- Floating point arithmetic
 - ✓ Addition
 - ✓ Subtraction
 - ✓ Multiplication

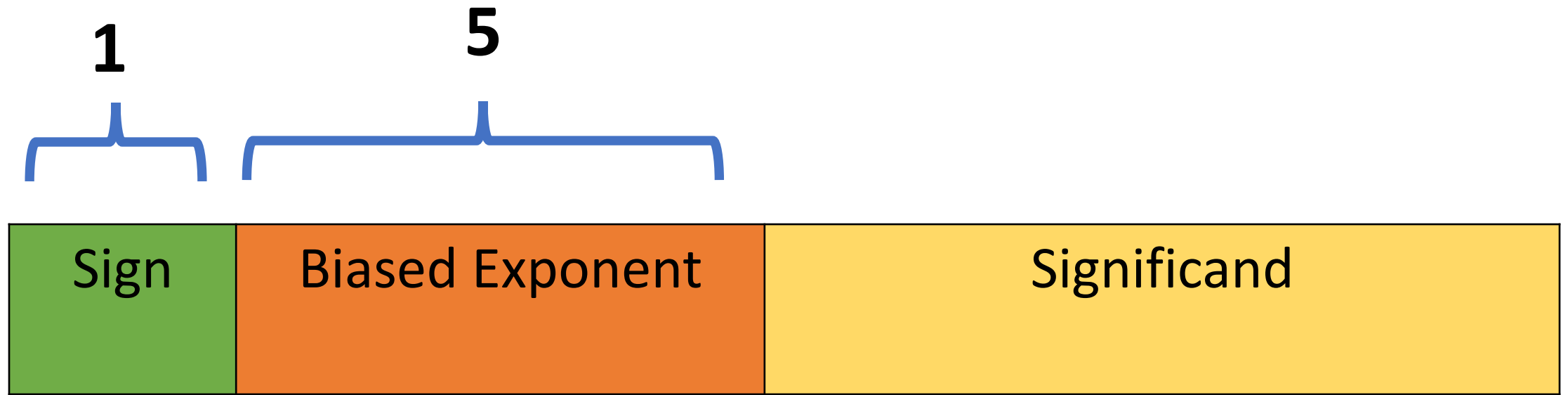
Floating point representation



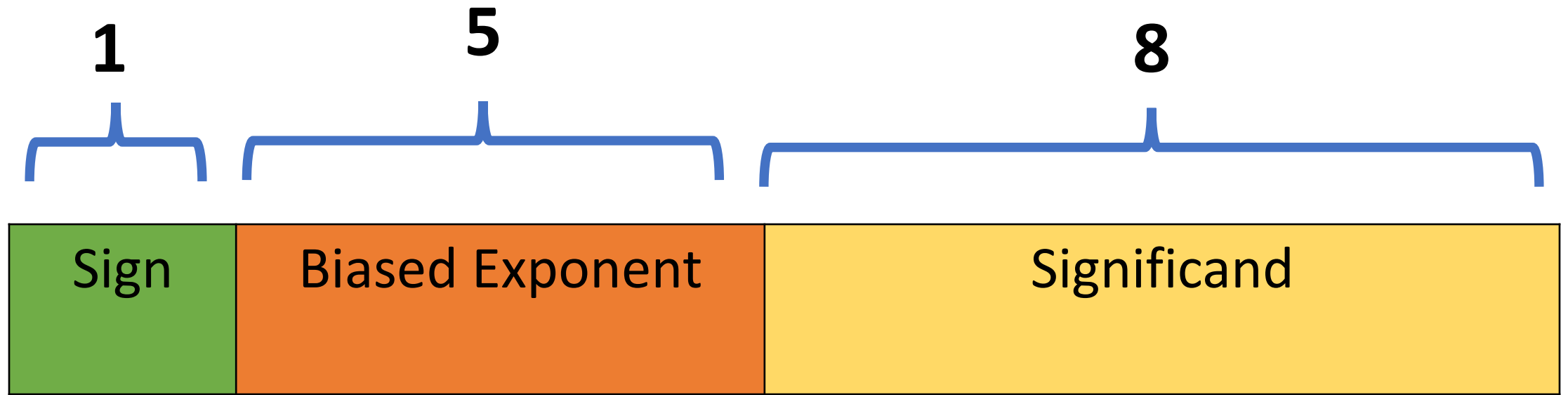
Floating point representation



Floating point representation

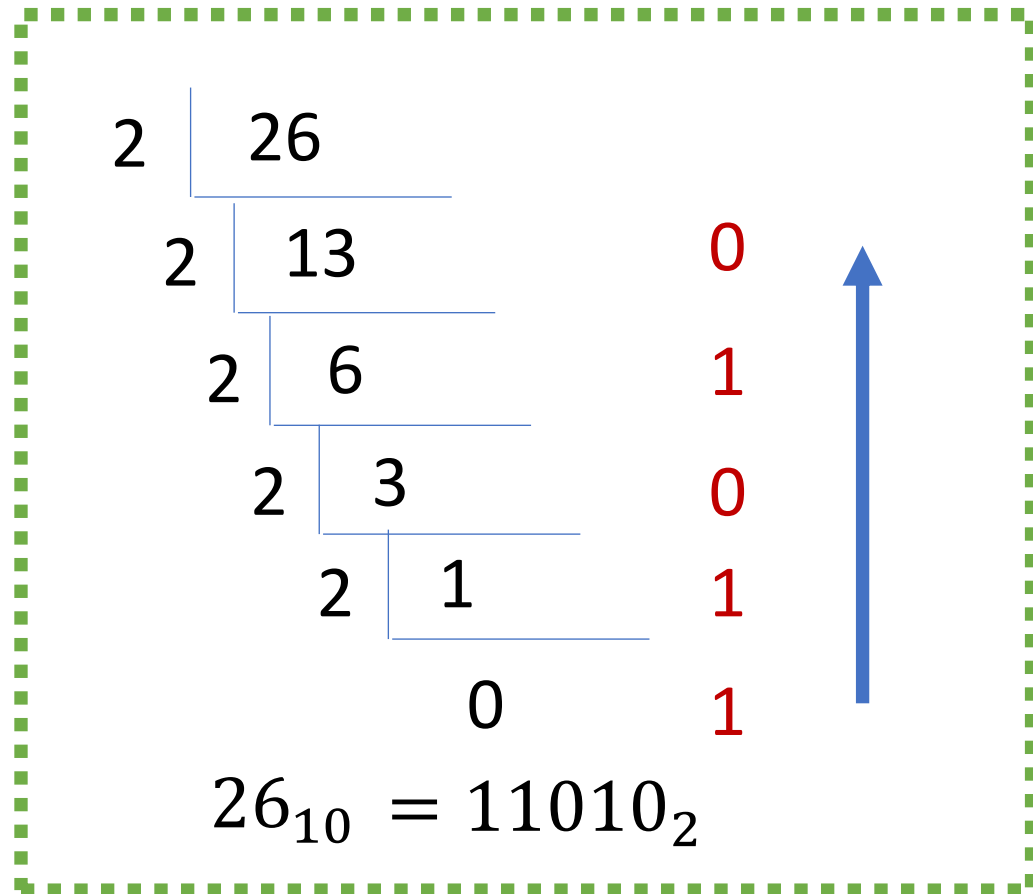


Floating point representation



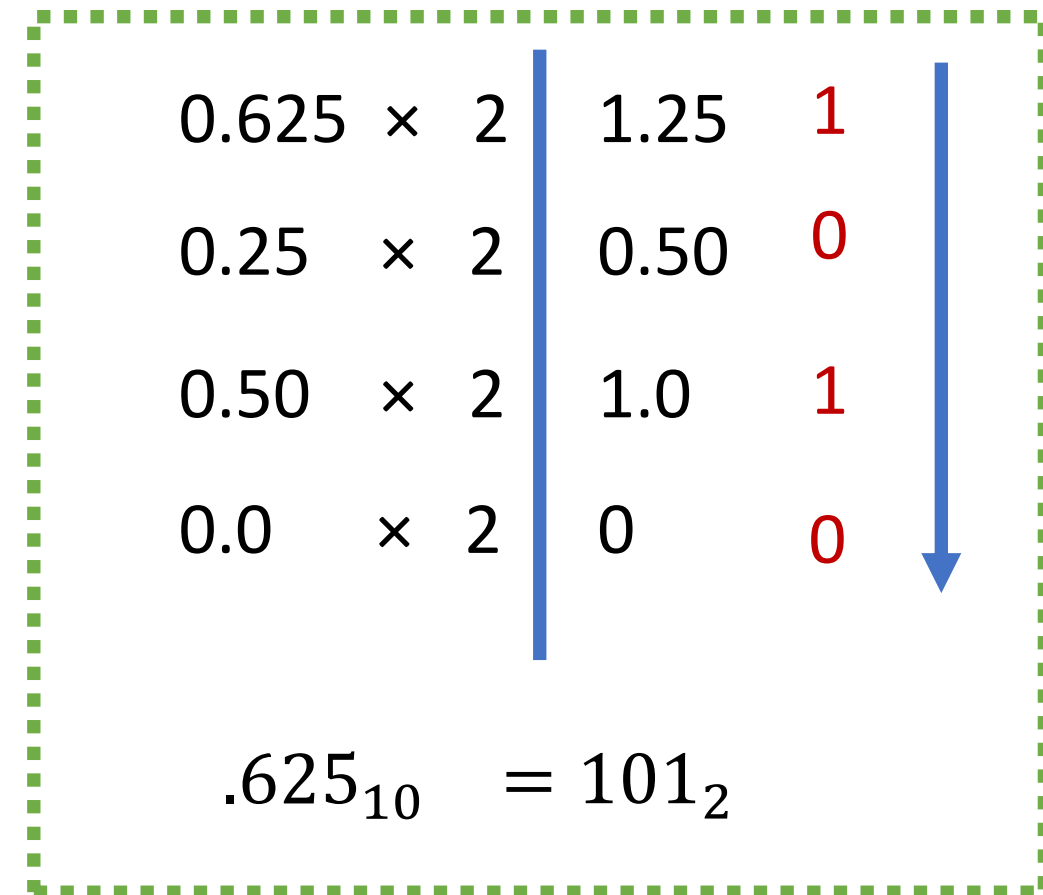
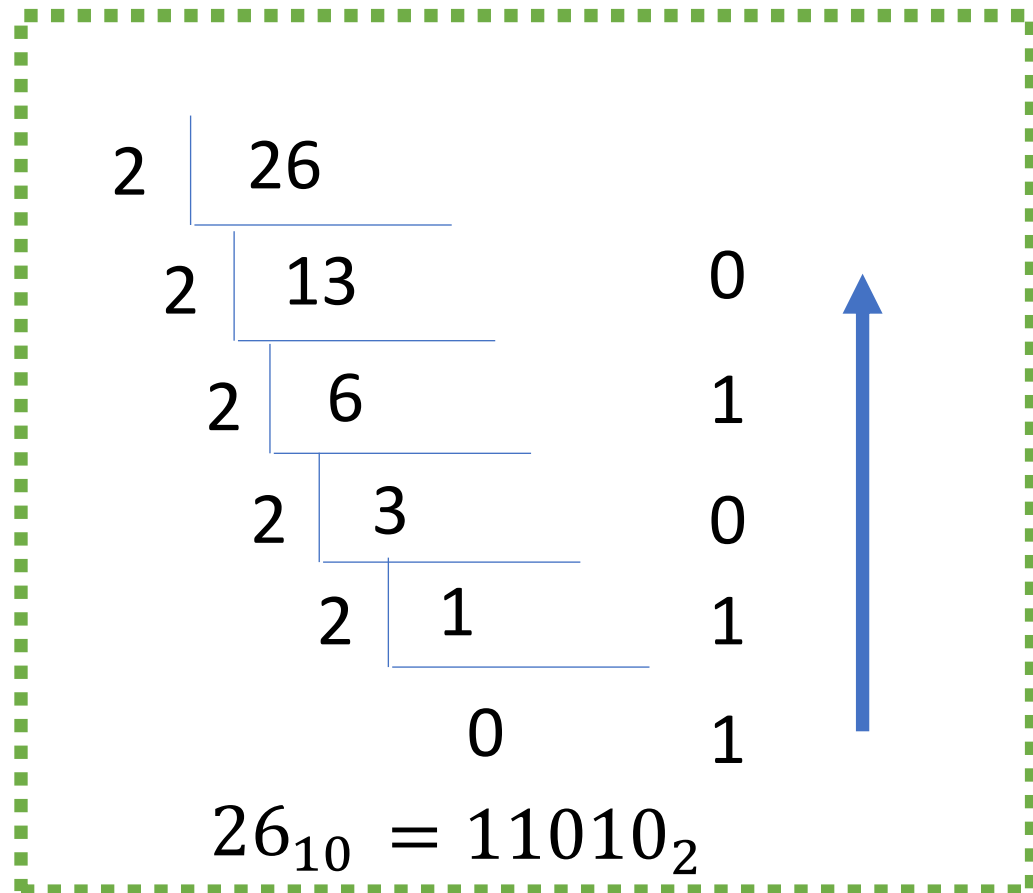
Decimal to binary

-26.625



Decimal to binary

-26.625



Floating point representation

Step 1: Binary representation

$$26.625_{10} = 11010.101_2$$

Floating point representation

Step 2: Normalization

$$11010.101_2 \Rightarrow 0.11010101 \times 2^5$$


*Significant
/Mantissa*

Floating point representation

Step 3: Calculate Bias exponent

N (5 in our model) is the number of bits allocated for biased exponent
Exp (5 in our example) is the power of 2 in our example.

- **Biased exponent** = $(2^N - 1) + \textit{Exp}$
= $(2^5 - 1) + 5$
= **20**
= **10100_2**

Floating point representation

-26.625_{10}

Step 4:

- Sign bit = 0 for positive , 1 for negative



Floating point representation

-26.625_{10}

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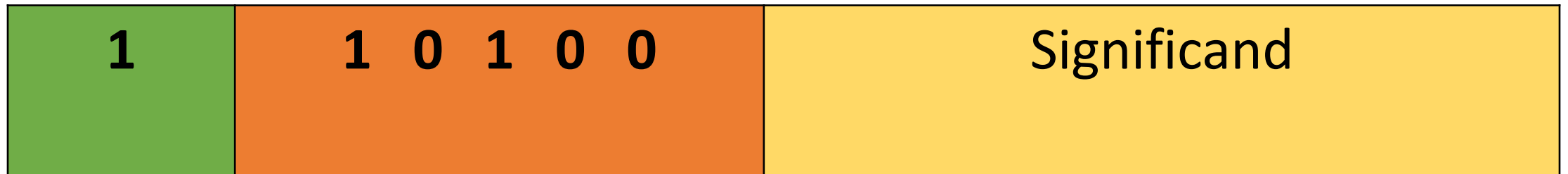


Floating point representation

-26.625_{10}

Step 4:

- Sign bit = 0 for positive , 1 for negative
- Place Biased exponent = 10100_2

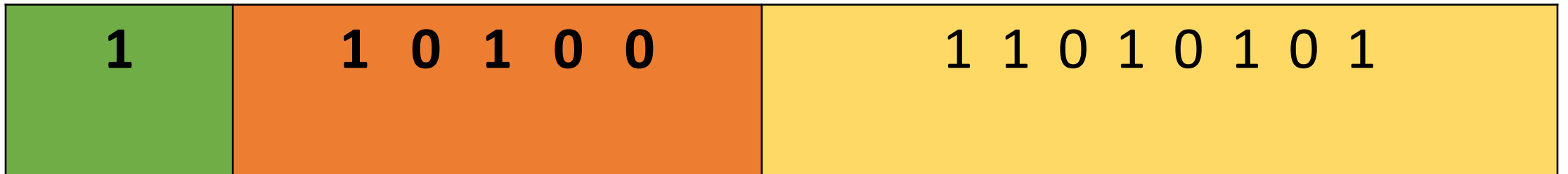


Floating point representation

-26.625_{10}

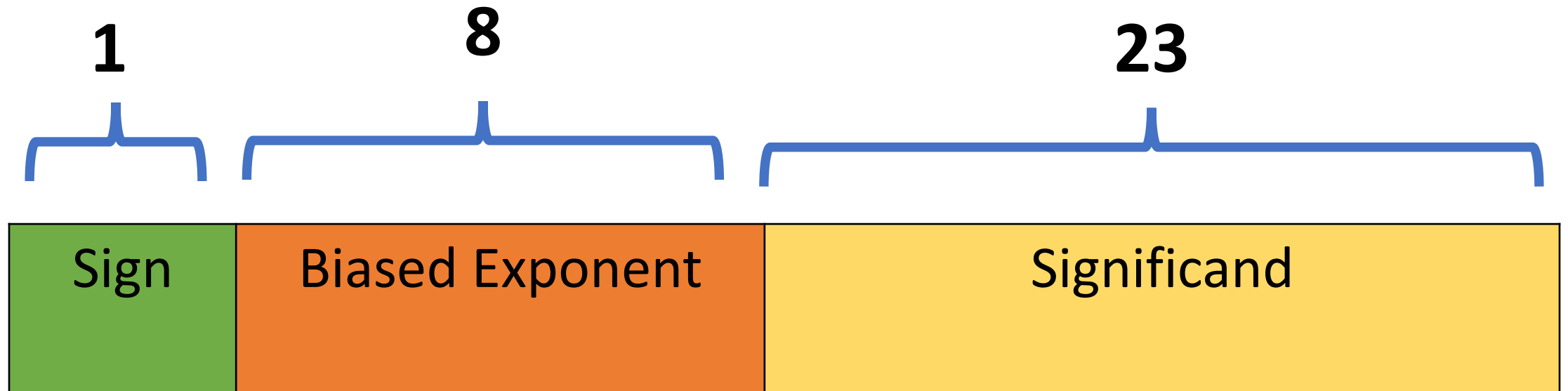
Step 4:

- Sign bit = 0 for positive , 1 for negative
- Place Biased exponent $= 10100_2$
- Place Significand $\Rightarrow 0.11010101 \times 2^5$



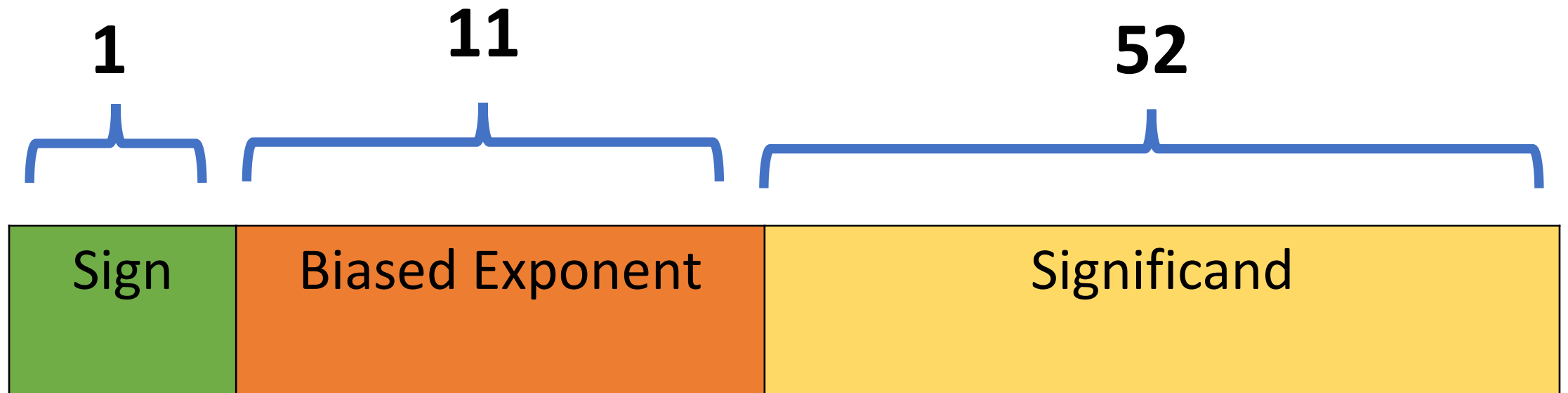
Floating point representation IEEE - 754

Single Precision



Floating point representation IEEE - 754

Double Precision



Single Precision

Floating point representation IEEE - 754

- Step 1: Decimal to binary conversion

For -26.625_{10}

$$26.625_{10} = 11010.101_2$$

Single Precision

Floating point representation IEEE - 754

Step 2: Normalization

$$11010.101_2 \Rightarrow 1.1010101 \times 2^4$$

*Significant
/Mantissa*

Floating point representation IEEE - 754

Step 3: Calculate Bias exponent

N (8 in our model) is the number of bits allocated for exponent
Exp (4 in our example) is the power of 2 in our example.

- **Biased exponent** = $(2^N - 1) + \text{Exp}$
= $(2^8 - 1) + 4$
= 131
= **10000011₂**

Floating point representation IEEE - 754

-26.625_{10}

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Floating point representation IEEE - 754

-26.625_{10}

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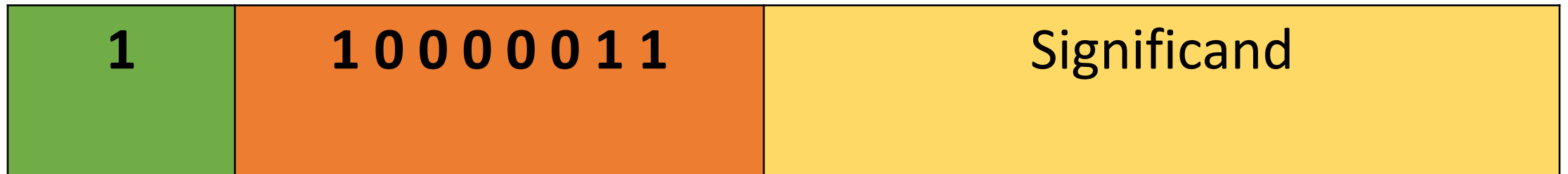


Floating point representation IEEE - 754

-26.625_{10}

Step 4:

- Sign bit = 0 for positive , 1 for negative
- Place Biased exponent = **10000011_2**

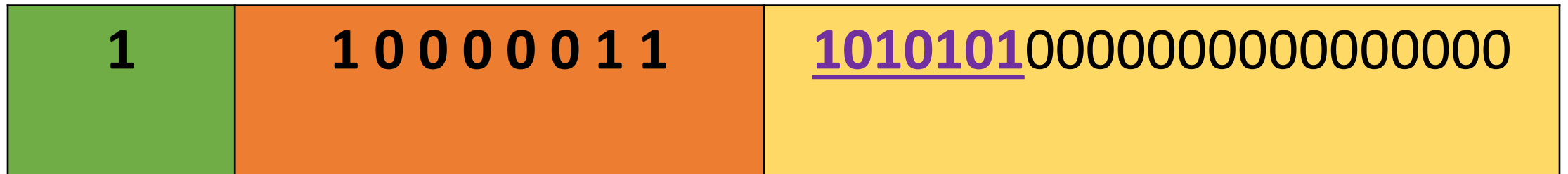


Floating point representation IEEE - 754

-26.625_{10}

Step 4:

- Sign bit = 0 for positive , 1 for negative
- Place Biased exponent = 10000011_2
- Place Significand $\Rightarrow 1.1010101 \times 2^4$



Floating-Point Number	Single-Precision Representation
1.0	0 01111111 000000000000000000000000
0.5	0 01111110 000000000000000000000000
19.5	0 10000011 001110000000000000000000
-3.75	1 10000000 111000000000000000000000
Zero	0 00000000 000000000000000000000000
$\pm$ Infinity	0/1 11111111 000000000000000000000000
NaN	0/1 11111111 any nonzero significand
Denormalized Number	0/1 00000000 any nonzero significand

Find the sum of 12_{10} and 1.25_{10} using the 14-bit floating-point model.

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- $1.25_{10} = 0.101 \times 2^1 = 0.000101 \times 2^4$

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- **$12_{10} = 0.1100 \times 2^4$**
- **$1.25_{10} = 0.101 \times 2^1 = 0.000101 \times 2^4$**

- **Sum = 0.110101×2^4**

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- $1.25_{10} = 0.101 \times 2^1 = 0.000101 \times 2^4$
- Sum = 0.110101×2^4

+

0	1 0 1 0 0	1 1 0 0 0 0 0 0
0	1 0 1 0 0	0 0 0 1 0 1 0 0
0	1 0 1 0 0	1 1 0 1 0 1 0 0

Find the product of 12_{10} and 1.25_{10} using the 14-bit floating-point model.

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- Product = 0.0111100×2^5

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 $= 0.111100 \times 2^4$

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- **$1.25_{10} = 0.101 \times 2^1$**
- Product = **0.0111100×2^5**
 $= 0.111100 \times 2^4$